



FINAL PROJECT TECHNICAL REPORT

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CS 112: COMPUTER PROGRAMMING FOR CS

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INTRODUCTION.

Electricity utilities and health care clinics both depend on systems that combine reliable data analysis with secure, role-based software to manage day-to-day operations. This project brings both domains into a single integrated academic exercise. The first component analyses electricity- grid - dataset modelled on the structure of Ghanaian electricity institutions such as Electricity Company of Ghana, the Ghana Grid Company and the Volta river Authority to understand asset distribution, network connectivity and structural resilience, The second and third components extend this analytical foundation into role based software applications: GridCareLite and an outage and maintenance management system for a utility, and ClinicCare-Lite, an administrative and communication system for an outpatient clinic.

Objectives: [1] clean, integrate, and analyses the grid dataset. [2] model the grid as a network to identify structurally important substations. [3] build GridCare-Lite's outage-to-resolution workflow. [4] build ClinicCare-Lite as a strictly non-diagnostic clinic administration tool. [5] apply continuous testing and responsible documentation throughout.

Approach: Work was divided into four roles, data engineering/ analysis, data analysis/BI, visualization and software architecture.

DATASET OVERVIEW.

Source: The grid data was generated using a seeded dataset generator script. To ensure reproducible results, the generator was run using random.seed[42].

Key Fields: The generated datasets consist of three CSV files: utilities.csv, substations.csv, and lines.csv.

Quality issues checked during data cleaning: missing values, duplicates records, invalid coordinates, inconsistent categorical labels, and orphaned lines/ utilities.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	SubstationName	Short Name	Region	Country	Latitude	Longitude	Voltage (kV)	Capacity (MVA)	Commissioned	Type	Status				
2	1 Achimota	Achimota	Greater Accra	Ghana	5.6085	-0.2193	11	6.4	2008	Distribution	Active				
3	2 Tema Sub	Tema	Greater Accra	Ghana	5.6596	-0.0226	330	48.5	1997	Transmission	Active				
4	3 Mallam Sub	Mallam	Greater Accra	Ghana	5.563	-0.2971	330	25.2	1979	Transmission	Active				
5	4 Legon Sub	Legon	Greater Accra	Ghana	5.6401	-0.1799	161	241.9	2009	Transmission	Active				
6	5 Kaneshie Sub	Kaneshie	Greater Accra	Ghana	5.5691	-0.2413	161	146	1970	Transmission	Active				
7	6 Aboadze J	Aboadze J	Greater Accra	Ghana	5.5921	-0.1439	161	52.2	2011	Transmission	Active				
8	7 Kumasi Central	Kumasi Central	Ashanti	Ghana	6.6856	-1.623	161	221.4	2018	Transmission	Active				

EXPLORATORY DATA ANALYSIS

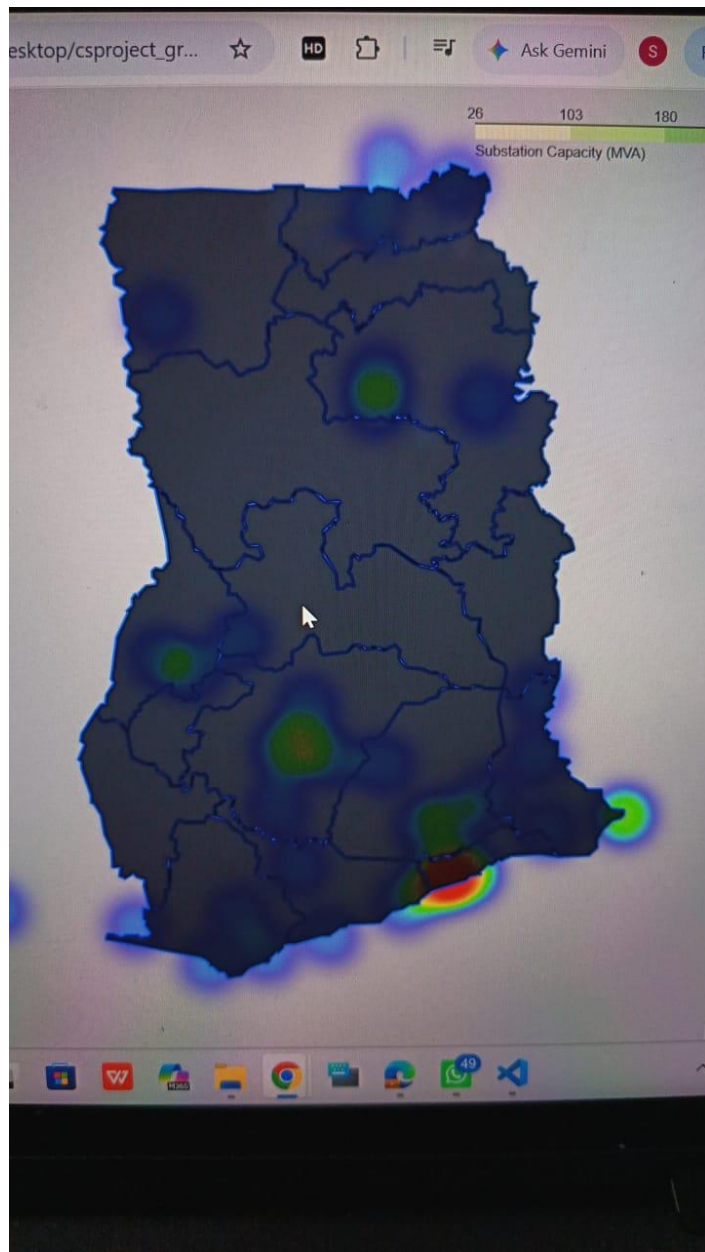
Key Visualizations: Substations cluster heavily in Greater Accra and Ashanti, with many border/northern regions having just one each. 161 kV and 330 kV are the most common voltage tiers. Capacity is right-skewed [mostly under 150 MVA, a few up to 487.6 MVA].

Geographically, larger/higher-voltage substations cluster in the south; the north and border areas are sparser and older. Lines are 96% Active, and mostly Overhead.

Statistical Insights: Capacity averages ~158 MVA [median ~109]. Infrastructure age shows a clear north-older, south-newer pattern. Transmission is the most common substation type. Mallam, Kumasi Central, and Cape Coast are the most-connected substations. No missing data is found.

NETWORK ANALYSIS.

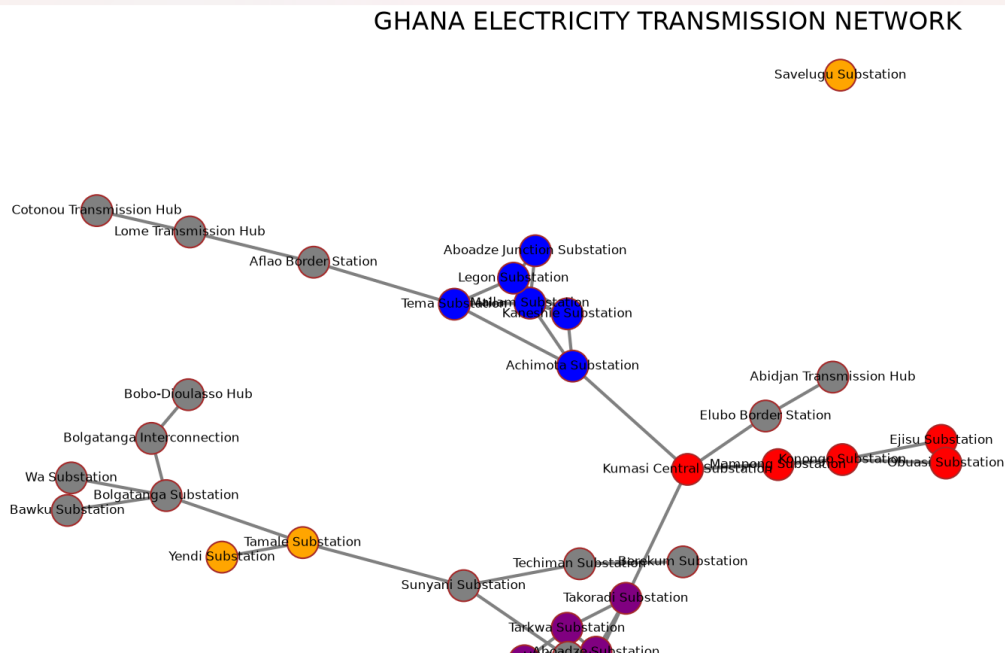
Graph Construction: The network was modeled as an undirected graph using NetworkX, with substations as nodes and transmission lines as edges. Metrics computed include betweenness, closeness centrality, PageRank, Diameter, average path length, clustering coefficients, and global efficiency.



Community detection grouped substations into clusters, compared against regional boundaries.

A simplified N-1 Contingency removed each top 5 critical substations individually and measured resulting fragmentation.

Figure 1



GRIDCARE-LITE ARCHITECTURE.

Data Model: The backend relies on a SQLite database structured to manage utilities, substations, outages, and work orders.

Roles & Workflow: The system features role-based access control [RBAC]. Engineers log outages, administrators assign work orders, technicians to update statuses, and customer service staff log complaints.

Security Controls: authentication required, role-based access at app and DB level: input validation with graceful error handling; no unhandled crashes on bad input or DB failure.

CLINCCARE-LITE ARCHITECTURE

Data model: users [clinicians, patients] tasks, submissions, reviews, messages, notifications, engagement log.

ClinicCare-Lite

Login

Email:

Password:

Don't have an account? [Register here](#)

Roles: Roles: Clinician [creates tasks, reviews submissions, records categorical outcomes, messages, announcements, aggregated analytics]; Patient [views tasks, submits files, reads feedback, private inbox/messaging, own engagement history].

Workflow: clinician assigns task with due date → patient uploads an approved file [.txt/.csv/.pdf, validated for type/size/ownership] → clinician records outcome [Pending / Reviewed–Normal / Needs Follow-up / Escalated] with notes → patient notified.

Security: bcrypt-hashed passwords, session-based role protection; validated file uploads stored per-patient; private 1:1 messaging with a persistent "not for emergencies" notice; no patient can view another patient's data; no leaderboards/ranking of patient engagement; analytics scoped by role.

Non-diagnostic scope statement: ClinicCare-Lite remains strictly an administrative and communication system. It does not diagnose, interpret symptoms, calculate disease risk, recommend treatment, or replace clinical judgement. Automated checks are limited to non-clinical field/format validation, and review outcomes reflect a clinician's manual judgement, not an automated diagnosis.

TESTING SUMMARY

Approach: continuous testing across the data pipeline and both apps; each test recorded objective, input, expected/actual outcome, pass/fail, defect, fix, and retest result.

Coverage: data types, missing values, duplicates, FK integrity, and reproducibility [pipeline]; auth, role access, invalid status transitions, duplicate records [GridCare-Lite]; weak passwords, unauthorized access, file-type/size limits, messaging privacy, diagnostic-scope violations [ClinicCare-Lite].

Key defect: an N-1 contingency loop referenced an undefined variable with an empty body [NameError]. Fixed by iterating over the existing critical-substation list, copying the graph per removal, and recomputing components. A missing scipy dependency [required by NetworkX PageRank] was also found and resolved.

DISCUSSION, CONCLUSION AND RECOMMENDATIONS

Working on CS 112 Computer Programming for CS Final Course Project Summer 2026 was both exciting and overwhelming. We were proud to see three different systems come together—from analyzing Ghana’s electricity grid with NetworkX to building GridCare-Lite and ClinicCare-Lite with secure, role-based workflows. At the same time, we struggled with setting up our GitHub account and how to use it, integrating datasets, fixing bugs, resolving Git conflicts, and coordinating our work under tight deadlines. There were moments when features broke unexpectedly, or different modules refused to work together, but those setbacks taught us patience, better communication, and the importance of testing before integration. More than anything, this project reminded us that great software is built through teamwork, persistence, and learning from mistakes, not getting everything right on the first try.

REFERENCES

Balibrea Mas, D. [2024]. Bcrypt: Modern password hashing for your software and your servers [Computer software]. <https://pypi.org/project/bcrypt>

Geopy contributors. [2024]. *geopy: Geocoding library for Python* [Computer software]. <https://geopy.readthedocs.io>.

Hagberg, A., Swart, P., & S Chult, D. [2008]. Exploring network structure, dynamics, and function using NetworkX. *Proceedings of the 7th Python in Science Conference [SciPy2008]*

Hunter, J. D. [2007]. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9[3], 90–95

McKinney, W. [2010]. Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 56–61

Pallets Projects. [2024a]. *Flask: A lightweight WSGI web application framework* [Computer software]. <https://flask.palletsprojects.com>

Pallets Projects. [2024b]. *Jinja: A fast, expressive, extensible templating engine* [Computer software]. <https://jinja.palletsprojects.com>

Plotly Technologies Inc. [2024]. *Plotly: Interactive graphing library for Python* [Computer software]. <https://plotly.com/python/>

Python Software Foundation. [2024]. *math — Mathematical functions* [Software documentation]. Python 3 documentation. <https://docs.python.org/3/library/math.html>

Python Software Foundation. [2024]. *Python language reference, version 3.x* [Computer software]. <https://www.python.org>

Python Software Foundation. [2024]. *sqlite3 — DB-API 2.0 interface for SQLite databases* [Software documentation]. Python 3 documentation. <https://docs.python.org/3/library/sqlite3.html>

SQLite Consortium. [2024]. *SQLite* [Computer software]. <https://www.sqlite.org>

Story, M., & Folium contributors. [2024]. *folium: Python data, leaflet.js maps* [Computer software]. <https://python-visualization.github.io/folium/>

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... van Mulbregt,

P. [2020]. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272.

World Wide Web Consortium [W3C]. [2024]. *HTML: HyperText Markup Language living standard*. <https://html.spec.whatwg.org>

Sowah, R. A. [2026]. *CS 112: Computer programming for CS — Final course project brief, Summer 2026* [Course document]. University of Ghana, Department of Computer Science.